Certified Correction of Vision–Language Hallucinations: Gains, Preconditions, and a Sequential Failure Mode

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Abstract

Selective prediction gives vision–language models (VLMs) a distribution-free guarantee on the answers they emit, at the cost of abstaining on the rest. A recent impossibility result shows this cost is irreducible: any predictor that may only *emit or abstain* the model's own answer must abstain on at least $(\mu - \alpha)/(1 - \alpha)$ of inputs when the base risk μ exceeds the target α. We observe that the bound's premise – that the emitted answer is the model's own – can be broken, and study what happens when a selective predictor is allowed to emit a *repaired* answer instead. We introduce **CCRC** (Certified-Correction Risk Control), which accepts the model's answer when a correctness score is high, substitutes an answer drawn from an *independent* evidence channel when that channel's evidence is strong, and abstains otherwise, retaining the guarantee $\Pr[\text{risk(emitted)} \leq \alpha] \geq 1 - \delta$. Three findings follow. (i) The nonconformity *score* dominates: a detection-grounded score raises certified coverage over model-internal uncertainty by up to 10.2 points, and over a faithful reproduction of a published CLIP-based scorer by far more — though we show by attribution that most of that margin is the score, not the procedure. (ii) Repairs must be certified by an independent channel; self-certification is provably circular and empirically dead. Where CCRC helps, a small mass of high-precision repairs ($\approx$1.5%) buys 3–6$\times$ its own size in coverage by diluting emitted risk, and we give a checkable precondition (μ comfortably above α) that also predicts the one dataset where CCRC loses. (iii) In the *generative* setting the natural sequential extension fails: in a paired matched-prefix experiment, repairing a claim significantly *increases* downstream hallucination ($+0.207 \pm 0.091$ mentions, $n = 121$, $p = 0.025$), which invalidates a fixed-point calibration argument and shows that myopic repair carries an unpriced downstream cost. We report negative results and attributions in full, and position the method as complementary to — not superseding — concurrent evidence-acquisition work.

1 Introduction

Vision–language models (VLMs) hallucinate: they assert objects, attributes and relations that the image does not support. A large body of work reduces hallucination heuristically, by contrastive or attention-modified decoding, external verification, or retrieval. These methods lower error rates but provide no *bound*: a deployer cannot state how often an emitted claim is wrong.

Conformal risk control supplies exactly that missing bound. Applied to VLMs, it yields *selective prediction*: emit an answer when a nonconformity score clears a calibrated threshold, abstain otherwise, and the error rate among emitted answers is provably at most α with probability $1-\delta$, without distributional assumptions. The price is abstention, and the price is not small. Anonymous (2026b) make this precise with a distribution-free lower bound: when the base risk μ exceeds α, *any* such predictor must abstain on at least $(\mu-\alpha)/(1-\alpha)$ of inputs, regardless of score quality, bound tightness, or calibration set size.

That bound rests on a premise which is easy to miss: the predictor may only *emit or abstain the model's own answer*. Every conformal method for VLMs we are aware of respects it — they delete claims, abstain, or re-score the original assertion. This paper asks what a selective predictor can do if it is allowed to emit a *different* answer.

Contributions.

- **CCRC**, a selective-prediction procedure that emits accepted, *repaired*, or abstained outputs and controls the risk of everything it emits (§4). Repairs are drawn and certified from an evidence channel that is independent of the score used to flag the original answer; we show this independence is not a design preference but a necessity (§4.2).
- An account of *why* it works and *when*. A small mass of high-precision repairs lowers the emitted set's average risk and thereby relaxes the binding constraint, buying 3–6× its own size in coverage. The gain tracks the abstention floor $\mu-\alpha$, giving a precondition that is checkable before any calibration and that correctly predicts the single dataset on which CCRC loses (§4.4).
- A **sequential negative result**. The natural generative extension — repair a hallucinated claim, keep decoding — is worse than it looks: in a paired matched-prefix experiment, repair significantly *increases* hallucination in the continuation. This rules out the monotone fixed-point argument that a sequential guarantee would otherwise rest on, and identifies an unpriced downstream cost that any correct sequential procedure must account for (§6).
- An **attribution analysis** of our own gains. Against a faithful reproduction of a published CLIP-based scorer, our total improvement is large, but we decompose it and report that most of it comes from the nonconformity score rather than from the procedure (§5). We also run the closest concurrent method and find the two are *complementary* rather than competing (§5).

We regard the negative results as part of the contribution rather than as caveats attached to it, and we have tried to make every claim in the paper checkable from the released code and extracted per-item signals.

2 Background and setup

Selective prediction with a risk guarantee. A VLM answers a query about an image. Let $Y \in \{0,1\}$ indicate whether the model's answer is correct, $\mu = \Pr[Y=0]$ the base risk, and $s(x) \in [0,1]$ a score intended to increase with correctness. A filtering policy emits when $s(x) \geq \tau$. Calibrating τ on a held-out set so that the error rate among emitted answers is at most α with confidence $1-\delta$ is a conformal risk control problem (Bates et al., 2021; Angelopoulos et al., 2021). We use exact Clopper–Pearson upper confidence bounds (Clopper & Pearson, 1934) on the emitted-set error rate, and fixed-sequence testing over a pre-specified nested family of policies, which controls the error probability without a multiplicity penalty.

Table 1: Selective prediction on POPE (LLaVA-1.5-7B, $n = 1500$). AURC is the area under the risk–coverage curve (lower is better); cov@10% is certified coverage at $\alpha = 0.10$. Realised test risk was $\leq \alpha$ for every row.

Score	AURC ↓	cov@10% ↑	AUROC ↑
Raw confidence	0.0775	62.1%	0.767
Learned, confidence only	0.0681	65.2%	0.786
+ CLIP grounding	0.0632	67.5%	0.800
+ detection grounding (OWLv2)	**0.0546**	**73.6%**	**0.835**
+ both	0.0538	74.9%	0.838

The abstention floor.

Proposition 1 (Anonymous 2026b, Prop. 3)**.** *If $\mu > \alpha$, any selective predictor that emits or abstains the model's own answer must abstain on at least $(\mu - \alpha)/(1 - \alpha)$ of inputs to guarantee emitted-set risk $\leq \alpha$, irrespective of score quality, bound tightness, or calibration size.*

The floor depends only on μ and α, both known in advance. It is the reason selective prediction is expensive on hard benchmarks, and it is the constraint the rest of this paper is organised around.

Two channels. Throughout, *channel 1* is a learned correctness score $s(x) = \hat{\Pr}[Y = 1 \mid \text{features}]$ built from the VLM's own confidence together with grounding features; *channel 2* is an open-vocabulary object detector that can answer the query itself, with evidence margin $m(x)$. Channel 2 is what makes repair possible, and its independence from channel 1 is what makes repair certifiable.

Protocol. Unless stated otherwise every number in this paper uses a three-way disjoint split (fit the score / calibrate the threshold / test), fixed-sequence testing with Clopper–Pearson at $\delta = 0.10$, and 20–100 random repetitions; we report the realised test risk alongside coverage so validity can be audited in every table. The three-way split is not incidental: §7 shows that reusing the calibration fold to fit the score silently breaks the guarantee for high-capacity scores.

3 The nonconformity score

Validity holds for *any* score; only *efficiency* — how much output survives at a given α — depends on which score is used. We therefore first ask what kind of score is efficient for VLM hallucination.

Table 1 compares scores on POPE with LLaVA-1.5-7B. A learned combination of the model's own confidence signals is a strong baseline. Adding generic image–text similarity (CLIP) helps a little; adding an open-vocabulary *detection* signal helps substantially. On the adversarial POPE split, and against a baseline that is given both confidence and multi-sample self-consistency — the scoring philosophy of Anonymous (2025b) — detection grounding still adds 10.2 ± 8.6 points of certified coverage.

Two observations shape the rest of the paper. First, the advantage of grounding *grows as the guarantee tightens*: at $\alpha = 0.05$ detection grounding nearly doubles usable coverage (30.0% → 52.5%), while at $\alpha = 0.20$ both scores certify almost everything. Second, the benefit tracks how much the base model hallucinates: on Qwen2-VL-2B, which is more accurate on this benchmark, the same addition is worth only 1.9 points. Grounding is most valuable exactly where selective prediction is most expensive.

Remark 1 (accuracy is not separability)**.** *A detector whose standalone accuracy matches the VLM's on POPE (*83.1% *vs.* 81.9%*) certifies far less coverage as a selective predictor (*55.3% *vs.* 68.2%*). Conformal coverage depends on how well a score* separates *correct from incorrect outputs, not on how often the underlying predictor is right.*

Table 2: CCRC (q=0.10) versus filtering, identical protocol. Realised test risk was $\leq \alpha$ in every row; the risk column is CCRC's.

Dataset	Backbone	n	μ	α	Filter	CCRC	Gain
POPE	LLaVA-1.5	1500	18.1%	0.10	68.2%	**72.6%**	+4.4
POPE-adv	LLaVA-1.5	444	17.3%	0.10	42.1%	**46.8%**	+4.7
POPE-adv	Qwen2-VL-2B	591	12.9%	0.10	77.5%	**79.4%**	+1.9
POPE-adv	LLaVA+VCD	591	19.6%	0.10	45.0%	**47.5%**	+2.5
POPE-adv	LLaVA-1.5	444	17.3%	0.15	69.2%	**77.2%**	+8.0
AMBER(d)	LLaVA-1.5	228	11.4%	0.10	92.1%	84.5%	−7.5

4 CCRC: certified correction

4.1 Procedure

Fix a strict, pre-specified evidence quantile q (we use $q = 0.10$ throughout: repair only in the top decile of channel-2 evidence). For a permissiveness parameter λ,

$$\begin{aligned} \textsc{accept} \text{ the model's answer} \quad & \text{if } s(x) \geq Q_s(1-\lambda), \\ \textsc{repair} \text{ with channel 2's answer} \quad & \text{if not accepted and } m(x) \geq Q_m(1-q), \\ \textsc{abstain} \quad & \text{otherwise.} \end{aligned}$$

The emitted set is the union of accepted and repaired items, and its error counts what is *actually emitted*:

$$\text{err} = \#\{\text{accepted} \wedge \text{model's answer wrong}\} + \#\{\text{repaired} \wedge \text{channel-2 answer wrong}\}.$$

Because q is fixed, the family is nested in λ: as λ grows the accepted set grows and the repair set shrinks. A single fixed-sequence test over this family with Clopper–Pearson bounds at level δ therefore certifies $\Pr[\text{risk(emitted)} \leq \alpha] \geq 1 - \delta$ with no multiplicity correction. Proposition 1 does not apply, since the emitted answer is not always the model's.

Remark 2 (a practical detail with teeth)**.** *Clopper–Pearson cannot certify risk $\leq \alpha$ with fewer than $k_{\min} = \ln\delta/\ln(1-\alpha) \approx 22$ clean observations. A fixed sequence that starts below $k_{\min}$ fails at its first step and aborts, returning zero coverage. Our grid starts at the analytically derived minimum.*

4.2 Independence is necessary, not preferred

The tempting design is to certify a repair with $1-s$: if the score says the answer is probably wrong, emit the opposite. This is circular, and it fails empirically. A hallucination score reports *suspicion*, not knowledge of the truth. On POPE, even in the bottom 2% of s the model's answer was still correct 13.3% of the time, exceeding $\alpha = 0.10$; no repair region is certifiable this way, and a first version of CCRC built on this idea repaired essentially nothing. The detector channel, by contrast, is 93–100% accurate in its top evidence deciles, so repairs drawn there are certifiable. Certified correction therefore *requires* a second, independent evidence source — which is a substantive constraint on system design, not a modelling convenience.

4.3 Why it gains: risk dilution

We initially expected repairs to consume the accepted set's unused risk budget. The measured effect is the opposite and stronger. Repairs admitted at a strict evidence gate are *more* reliable than the acceptance gate's *marginal* items, so adding them *lowers* the emitted set's average risk. That relaxes the binding constraint and lets the acceptance gate open further along the nested family. The result is leverage: with 0.9–1.8% of items repaired, certified coverage rises by 1.9–4.7 points, i.e. 3–6× the repaired mass (Table 2).

Table 3: Attribution on POPE ($\alpha = 0.10$). Most of the margin over a published scorer comes from the score, not from our procedure.

Method	Coverage	Δ
CLIP-similarity scorer (Anonymous, 2025b), filtering	5.6%	—
+ VLM confidence	41.5%	+35.9
+ detection grounding	68.2%	+26.7
+ certified correction (CCRC)	**72.6%**	+4.4

Table 4: One common base score, one mechanism added per arm (POPE, LLaVA-1.5, $n = 394$). All arms valid.

Arm	cov@0.10	cov@0.15
Base filter (confidence only)	5.5%	16.5%
+ evidence acquisition (Anonymous, 2026a)	5.5%	24.1%
+ detection-grounded score (ours)	32.1%	63.2%
+ certified correction (ours)	34.5%	71.8%
Composed	**35.8%**	**74.2%**

4.4 When it helps: a checkable precondition

The last row of Table 2 is a loss. We isolated its cause with two controlled tests. It is *not* a weak repair channel: the detector is 95.6% accurate on that dataset and 100% accurate in the region repairs are drawn from. It is *not* small sample size: subsampling POPE to the same n still gains +3.2. The cause is missing headroom. With $\mu = 11.4\%$ barely above $\alpha = 10\%$, the abstention floor of Proposition 1 is only 1.6%; filtering alone already certifies 92.1%, so there is nothing for repair to recover, and admitting repaired mass perturbs a nearly-saturated constraint.

The gain therefore tracks $\mu - \alpha$: it is +1.9 to +4.7 points when $\mu - \alpha \gtrsim 3$ points and -7.5 when $\mu - \alpha \approx 1.4$ points. Both quantities are known before any calibration, so a practitioner can decide whether to use CCRC *a priori*. CCRC is a targeted tool for the high-base-risk regime — precisely the regime in which Proposition 1 forces heavy abstention — and not a drop-in improvement.

5 Attribution and comparison

Against a published scorer. Table 3 places the CLIP-similarity scoring function of Anonymous (2025b) in our pipeline and then adds our components one at a time. The total improvement is large, but the decomposition matters: most of it comes from the *score*, and only 4.4 points from certified correction. We state this explicitly because the aggregate number alone would misattribute the gain. This comparison also understates Anonymous (2025b), whose method was designed for free-form caption claims rather than yes/no probes; we transplant their scorer, not their system.

Against the closest concurrent method. Anonymous (2026a) rescue borderline claims by *acquiring more visual evidence* (zoomed views) and re-scoring the model's original assertion; they explicitly do not change the answer. Their machinery (Clopper–Pearson with fixed-sequence testing), domain, and goal coincide with ours, which makes them the right comparison. Holding the base score fixed and adding one mechanism per arm (Table 4), the two mechanisms are **complementary**: composing them beats every single-mechanism arm at both risk levels. Re-reading the image and replacing the answer rescue different claims. We do not claim to beat Anonymous (2026a): our reproduction uses only three coarse crops and under-delivers relative to their published gain, so the defensible conclusion is composition, not superiority.

Table 5: Paired matched-prefix test of monotonicity ($n = 121$). Repair significantly increases hallucination in the continuation.

Downstream measure	Original prefix	Repaired prefix
Hallucinated mentions (mean)	1.207	**1.413**
Per 100 words	3.61	4.30
Faithful mentions (mean)	1.579	1.752
Paired difference $+0.207 \pm 0.091$; $t = 2.27$, $p = 0.025$; Wilcoxon $p = 0.035$		
Better / worse / tie: 25 / 45 / 51		

6 A sequential failure mode

Extending certified correction from single answers to free-form generation looks straightforward: decompose a caption into claims, repair the unsupported ones, continue decoding. Two obstacles are known — editing a prefix makes later tokens off-policy, an instance of feedback covariate shift (Fannjiang et al., 2022), and per-claim guarantees say nothing about a whole sequence. A guarantee could be recovered by calibrating *on-policy*: iterate calibration and deployment to a fixed point. That argument needs the correction map to be monotone, so we measured whether it is, before attempting a proof.

Design. For each image we generate a caption greedily, locate the first hallucinated object mention at position t, and construct two prefixes that are *identical up to* t: one retaining the hallucinated word, one substituting the best-detected object that the image genuinely contains. We then continue decoding equally from each and count hallucinated mentions in the continuations only. Matched prefixes make the comparison causal.

Result. Repair makes the continuation *worse* (Table 5). Downstream hallucinated mentions rise from 1.207 to 1.413, a paired difference of $+0.207 \pm 0.091$ ($n = 121$, $t = 2.27$, $p = 0.025$; Wilcoxon $p = 0.035$); 45 cases worsen against 25 that improve.

Consequences. The monotone fixed-point argument is unavailable: the map is not monotone, so the iteration has no guaranteed fixed point and that proof should not be attempted. A guarantee on corrected generations must instead be obtained by coupling, bounding $\text{risk}(\text{corrected}) \leq \text{risk}(\text{original}) + \Pr[\text{repair wrong}]$ and controlling both terms. More importantly, the result identifies a cost that a myopic procedure does not price: greedy per-claim repair optimises local risk while incurring roughly 0.21 additional hallucinated claims downstream. Any correct sequential procedure must price the continuation rather than the claim.

Scope. Our repair substitutes the globally best-detected object, which can be contextually inappropriate mid-sentence and may push the prefix off-policy. The supported claim is that *context-blind* substitution carries a downstream cost, not that all correction must. Constraining repairs to the model's own top-k continuations is a natural remedy and remains to be tested.

7 Ablations

The combiner, and a validity trap. Higher-capacity score combiners do not help, and they can silently invalidate the guarantee. When the combiner is fitted on the fold used to calibrate τ, gradient boosting and random forests appear to certify far more coverage (91.3% and 88.5%) but their realised test risk is 13.3% and 12.1% against a 10% target: they overfit the calibration scores, so the threshold is set too permissively. Under a three-way split all combiners are statistically tied ($\approx$ 72–76%) and all are valid, while a small multilayer perceptron collapses ($\approx$ 28%, high variance). We therefore use logistic regression, and we fit the score on a fold disjoint from calibration. The features are a handful of informative scalars; capacity is not the bottleneck.

The risk bound. The emitted-set error is a binomial proportion, so an exact interval dominates general concentration inequalities. At $\alpha = 0.10$, Clopper–Pearson certifies 72.5% coverage, empirical Bernstein 54.9%, and Hoeffding 38.3%; all three are valid, and the latter two are simply conservative. Variance-adaptive or betting-style bounds become preferable for graded (non-binary) losses.

8 Related work

Hallucination mitigation. Decoding-time and verification-based methods reduce VLM hallucination without a guarantee: over-trust penalties (Huang et al., 2024), visual contrastive decoding (Leng et al., 2024), attention-based localisation and intervention (Anonymous, 2025a), and generate-then-verify pipelines (Anonymous, 2025c). These operate at full coverage and are complementary to risk control; we show in the appendix-level experiments that our layer composes on top of one of them (visual contrastive decoding), improving its selective behaviour substantially.

Conformal risk control for language and vision–language models. Conformal prediction has been adapted to language generation via calibrated sampling, stopping and rejection rules (Quach et al., 2024), to claim-level factuality (Anonymous, 2025b), and to abstention. Anonymous (2026b) characterise when certification is possible at all and supply Proposition 1. All of these filter, abstain, or re-score; none emits a modified answer, which is the gap this paper addresses. Anonymous (2026a) is closest in spirit and setting: it also seeks to avoid heavy abstention using additional visual evidence, but rescues claims by re-scoring rather than replacing them, and we find the two mechanisms compose (§5).

Distribution shift induced by the predictor. Editing a generation makes subsequent tokens off-policy. This is feedback covariate shift rather than exogenous shift, since the shift depends on the deployed threshold (Fannjiang et al., 2022). Our sequential result (§6) shows that in this setting the shift is not benign.

9 Limitations

Our gains are modest in absolute terms (+1.9 to +8.0 points of certified coverage) and one dataset shows a loss, explained by the precondition in §4.4. The repair gate q is a hyperparameter fixed *a priori* at 0.10 rather than tuned; loose gates can lose substantially. Channel 2 is an object detector, so the method as evaluated covers object-existence hallucination; on benchmarks dominated by reasoning or attribute questions the detector does not apply and the procedure reduces to filtering (which remains valid, and correctly abstains almost entirely on a near-chance benchmark). Channel-2 independence is an assumption; a detector sharing failure modes with the VLM would weaken, though not invalidate, certification. Finally, the one-shot guarantee is a correct application of existing machinery rather than new theory, and the sequential guarantee is not established — §6 shows the most natural route to it is closed.

10 Conclusion

Selective prediction for VLMs is expensive because of a bound that assumes the emitted answer is the model's own. Relaxing that assumption is worthwhile but not free: certified correction recovers coverage when the base risk is comfortably above the target, requires an independent evidence channel, and works through a dilution effect that gives it leverage over the repaired mass. In the generative setting the same idea, applied myopically, makes things worse in a way that is measurable and that constrains how a sequential version can be built. We hope the negative results are as useful as the positive ones.

Broader Impact Statement

Methods that attach guarantees to model outputs can improve safety, but a guarantee is only as meaningful as its assumptions. Our procedure controls the error rate among emitted answers under exchangeability and requires a genuinely independent evidence channel; if either fails — for instance under distribution shift, or with a detector that shares the VLM's blind spots — the stated risk level may not hold in deployment.

Emitting a *corrected* answer also changes the failure mode presented to a user, from an abstention that invites scrutiny to a confident substitution that may not. We therefore regard the checkable precondition and the sequential negative result as safety-relevant parts of the contribution rather than as limitations to be minimised.

Acknowledgments

Placeholder.

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